

Benjamin Johnston

Systems Engineering · Automation · SDK Design

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EXPERIENCE

AI Operations Engineer (RPA & Model Evaluation)

Nov 2025 – Jun 2026

Handshake — Contract

Columbus, OH

- Cut manual annotation and triage effort by 95% across 400+ contractor hours by building AutoHandshake!, a Ruby/JavaScript Selenium automation suite with UI-stall recovery, multi-tab orchestration, and full audit trails.
- Evaluated 4,000+ multimodal AI responses against structured quality rubrics for correctness, reasoning accuracy, and domain understanding, earning the Handshake Model Validation 2 — Expert credential.
- Mentored contractors on platform workflows and rubric interpretation, raising evaluation consistency team-wide and reducing rework from misaligned quality standards.

Expert Contributor

Dec 2025 – May 2026

Snorkel AI — Contract

Columbus, OH

- Engineered constraint-based training workflows for Claude and ChatGPT across 2 contracted projects targeting complex programming and logic reasoning tasks.
- Designed rigorous evaluation frameworks to systematically assess model reasoning capabilities, driving continuous improvements in code generation and logical deduction accuracy.
- Validated training-set integrity through Docker-based CI/CD, catching defects pre-release and protecting downstream model quality.

Search Quality Evaluator

Apr 2025 – Aug 2025

RWS Group — Contract

Columbus, OH

- Assessed search relevance and LLM output quality against detailed rubrics across Google query simulations and third-party models, identifying failure patterns that informed ranking and UX improvements.

Freelance Software Engineer

Jul 2021 – Present

Independent client engagements

Columbus, OH

- Architected and delivered SDKs, automation tooling, and API integrations for individual clients, including an AI-assisted grading platform with LMS OAuth that reached ~95% grading accuracy and cut instructor workload in half.
- Designed and maintain 10+ public open-source libraries across C#, C, C++, and D, spanning systems programming, protocol implementation, and developer tooling with ~53,000 lines of code and 700+ commits.
- Eliminated a user-mode virtual-input layer from a device-integration pipeline, reducing its overhead by 100% and removing it from the execution path entirely.

Biochemistry Intern

2022 – 2024

Columbus Alternative High School

Columbus, OH

- Managed safety data sheets (SDS/MSDS) and lab compliance documentation across ~12 science classes, ensuring all materials met district and state safety standards.
- Prepared lab materials and set up experiments for chemistry and biology courses, supporting hands-on instruction for hundreds of students across multiple grade levels.
- Graded assignments, lab reports, and exams using structured rubrics, applying consistent quality standards that paralleled later work in AI response evaluation.

SELECTED PROJECTS

UndeadInterop | C# | github.com/cetio/UndeadInterop 2024

- Built a Windows syscall research library in C# that resolves syscall IDs from PE export tables and invokes them directly via runtime-generated x86_64 shellcode stubs, bypassing the ntdll usermode layer entirely.
- Implemented 8 hook detection classifications (inline jumps/calls, stack pivots, debug interrupts, forwarded exports, tamper detection) by disassembling function prologues with the Iced x86 decoder; includes a complete NTSTATUS enum (339 values).

VEXAPI | C/C++ | github.com/cetio/VEXAPI 2024

- Reverse-engineered the VEX Robotics V5 private firmware API from ARM64 disassembly of VEXos using Ghidra, producing 8,650 lines of documented C/C++ headers across 39 files with 481 firmware memory offsets.
- Documented 34 public API headers covering 20+ device types (motor, IMU, vision, optical, display) and 5 private headers with jump table and graphics signatures; includes PoC for autonomous mode override.

Cassowary | C# | github.com/cetio/Cassowary 2024

- Authored a .NET runtime introspection library (7,400+ lines, 54 files) exposing CoreCLR internals as first-class managed types for reflection, allocation, and layout inspection without native code or P/Invoke.
- Modeled 15 CLR internal structures (MethodTable, EEClass, MethodDesc, FieldDesc) and 15 generic type signature models, providing an alternative to Microsoft's SOS debugging extension.

Nimbus | C | github.com/cetio/nimbus 2025

- Built a Linux CLI (2,500+ lines) for unified display configuration controlling brightness, contrast, gamma, and color temperature across both sysfs backlight and DDC/CI over I2C backends.
- Implemented function-pointer-based polymorphic display abstraction with runtime capability detection, supporting 5 VCP feature labels plus arbitrary hex codes; includes Ruby integration test suite.

Selenium | D | github.com/cetio/selenium 2026 – Present

- Authored the only maintained W3C-compliant Selenium WebDriver client for D (4,800+ lines, 28 modules), with typed APIs, driver auto-detection for 4 browsers, and full session lifecycle management.
- Implemented a Bridge/Driver architecture with concurrent session support; published on the DUB package registry with Selenium Grid support.

Intuit | D | github.com/cetio/intuit 2025 – Present

- Authored the first native LLM SDK for D (8,400+ lines, 67 modules), mapping chat completions, structured output, SSE streaming, and tool calls onto typed D structures.
- Supports OpenAI, Claude, and Qwen with OpenAI-compatible routing, SIMD-accelerated embeddings, and automatic JSON schema generation for type-safe native tool use.

TECHNICAL SKILLS

Systems Programming C, C++, C#, x86 Assembly, D

Scripting & Automation Python, Ruby, JavaScript, SQL

Protocols & APIs W3C WebDriver, OAuth 2.0, REST, DDC/CI, SSE streaming, JSON Schema, LLM integration

Infrastructure & Tools Linux, Git, CI/CD, Docker, CMake, Selenium, SIMD, HTML parsing

EDUCATION

Columbus State Community College

AAS, Software Development

Expected Dec 2026

Columbus, OH

CERTIFICATIONS

Handshake Model Validation 2 — Expert AI response evaluation across 4,000+ multimodal samples